

SOFTWARE ENGINEERING LAB – 2

(UE24CS341A)

Agile Backlog Creation & Sprint Simulation in Jira

Team Number: 007

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Problem Statement 7: Hostel Maintenance & Issue Ticketing System

EPIC: Epic 4 – Resolution Verification & System Reliability (FR-004 and NFR-002)

Note: This report covers only Epic 4, my individual contribution from the Lab 1 report. The remaining requirements are covered separately by my teammates: Epic 1 – FR-001 & FR-005: P. Saanvi; Epic 2 – FR-002 & NFR-001: Sonia Sharma; Epic 3 – FR-003: Kaustubh Akash.

Lab Objective

This lab converts the requirements FR-004 and NFR-002 from the Lab 1 report into an Agile Epic and User Stories in Jira, estimates effort using the Fibonacci scale, prioritises the backlog, simulates two one-week sprints on a Scrum board, and tracks sprint progress using the Burndown Chart.

1. Epic (Derived from FR-004 and NFR-002)

The two requirements are grouped into Epic 4: FR-004 mandates resident satisfaction by requiring a digital sign-off before a resolved ticket is permanently closed, while NFR-002 ensures platform resilience with a 99.9% uptime target (≤ 43 minutes downtime/month) throughout the semester.

Epic Key	Epic Name	Mapped Requirements	Description
HMT-20	Resolution Verification & System Reliability	FR-004 and NFR-002	Enable residents to inspect and digitally sign off on repairs before closure, and implement high-availability safeguards guaranteeing 99.9% system uptime.

2. User Stories for Epic 4

The Epic is broken into 5 User Stories using the format: As a [role], I want [goal], so that [benefit].

Key	Story	As a...	I want...	So that...	Covers
HMT-21	4.1 Resolution Notification & Inspection	Hostel Resident	to receive an alert when my ticket is marked 'Resolved' with work notes	I can inspect the physical repair in my room	FR-004
HMT-22	4.2 Digital Sign-Off Closure	Hostel Resident	to click a digital sign-off button confirming satisfaction	the ticket formally moves to 'Closed'	FR-004

HMT-23	4.3 Reopen Ticket with Feedback	Hostel Resident	to reject resolution and reopen the ticket with notes if the problem persists	the technician is sent back to fix it properly	FR-004
HMT-24	4.4 Inactivity Auto-Closure Policy	Maintenance Warden	unconfirmed tickets to auto-close after 48 hours of resident inactivity	resolved queues do not remain open indefinitely	FR-004
HMT-25	4.5 High Availability Watchdog	System Administrator	health checks and auto-restart policies guaranteeing 99.9% uptime	students can report emergencies anytime	NFR-002

Acceptance criteria were taken from Lab 1. FR-004: Ticket cannot be closed by staff alone; resident sign-off immediately updates status to Closed and archives ticket. NFR-002: System uptime logs across simulated stress tests show total downtime <= 43 minutes/month (99.9% availability) with automatic process recovery within 30 seconds of failure.

3. Jira Backlog – Epic, User Stories and Story Points

The screenshot shows Epic 4 with all 5 User Stories, the priority icon and story points on each row, and the Epic panel reporting 5 work items and an estimate of 24 points.

The screenshot displays the Jira Backlog interface. On the left, the 'Epic' panel for 'Epic 4 – Resolution Verification & System Reliability' shows 5 work items and an estimate of 24 points. The main backlog area lists the following user stories:

Story ID	Story Title	Priority	Story Points
HMT-10	4.1 Resolution Notification & Inspection	High	5
HMT-12	4.2 Digital Sign-Off Closure	High	5
HMT-13	4.3 Reopen Ticket with Feedback	Medium	3
HMT-14	4.4 Inactivity Auto-Closure Policy	Medium	3
HMT-15	4.5 High Availability Watchdog	Low	8

The total estimate for the epic is 24 points. The interface also shows a 'Sprint' panel for 'HMT Sprint 3' with 0 work items visible and an estimate of 0 points.

Figure 1: Jira Backlog – Epic 4 (HMT-20) with stories (5, 5, 3, 3, 8 = 24 pts).

4. Backlog Prioritization

User Stories were ordered by business value and dependency. The backlog order in Jira is 4.1, 4.2, 4.3, 4.4, 4.5.

Key	Story	Priority	Story Points
HMT-21	4.1 Resolution Notification & Inspection	High	5
HMT-22	4.2 Digital Sign-Off Closure	High	5
HMT-23	4.3 Reopen Ticket with Feedback	Medium	3
HMT-24	4.4 Inactivity Auto-Closure Policy	Medium	3
HMT-25	4.5 High Availability Watchdog	Medium	8

- 4.1 and 4.2 (High): Core sign-off requirement for FR-004; without them, resolved tickets can never reach final Closed state.
- 4.3 (Medium): Dispute handling feedback loop allowing residents to reject substandard repairs.
- 4.4 (Medium): Administrative safety net auto-closing abandoned tickets after 48 hours.
- 4.5 (Medium, 8 pts): Architectural infrastructure watchdog enforcing 99.9% availability.

Total Story Points (Epic 4): 24 | High: 10 | Medium: 14

5. Story Point Estimation – Planning Poker

Story points were estimated independently by each team member using the Fibonacci scale (1, 2, 3, 5, 8, 13...). The individual estimates were compared, differences were discussed, and the team agreed on the final estimates.

Story ID	Story Title	Saanvi	Sonia	Kaustubh	Kasissnu	Agreed
HMT-21 / 4.1	Resolution Notification & Inspection	5	5	5	5	5
HMT-22 / 4.2	Digital Sign-Off Closure	5	5	5	5	5
HMT-23 / 4.3	Reopen Ticket with Feedback	3	3	3	3	3
HMT-24 / 4.4	Inactivity Auto-Closure Policy	3	3	3	5	3
HMT-25 / 4.5	High Availability Watchdog	8	8	8	8	8

6. Sprint Planning & Simulation

Sprint 1 (Duration: 1 week)

Sprint Goal: Deliver resident inspection alerts and digital sign-off closure mechanism.

Key	Story	Priority	Pts	Status
HMT-21	4.1 Resolution Notification & Inspection	High	5	Done
HMT-22	4.2 Digital Sign-Off Closure	High	5	Done

Sprint 1 Velocity: 10 pts committed | 10 pts completed | 0 pts rolled over

Sprint 2 (Duration: 1 week)

Sprint Goal: Support issue reopening, 48-hour auto-closure, and 99.9% high availability infrastructure.

Key	Story	Priority	Pts	Status
HMT-23	4.3 Reopen Ticket with Feedback	Medium	3	Done
HMT-24	4.4 Inactivity Auto-Closure Policy	Medium	3	Done
HMT-25	4.5 High Availability Watchdog	Medium	8	Done

Sprint 2 Velocity: 14 pts committed | 14 pts completed | 0 pts rolled over

Jira Sprint Board (Active Sprint view) – Sprint 1

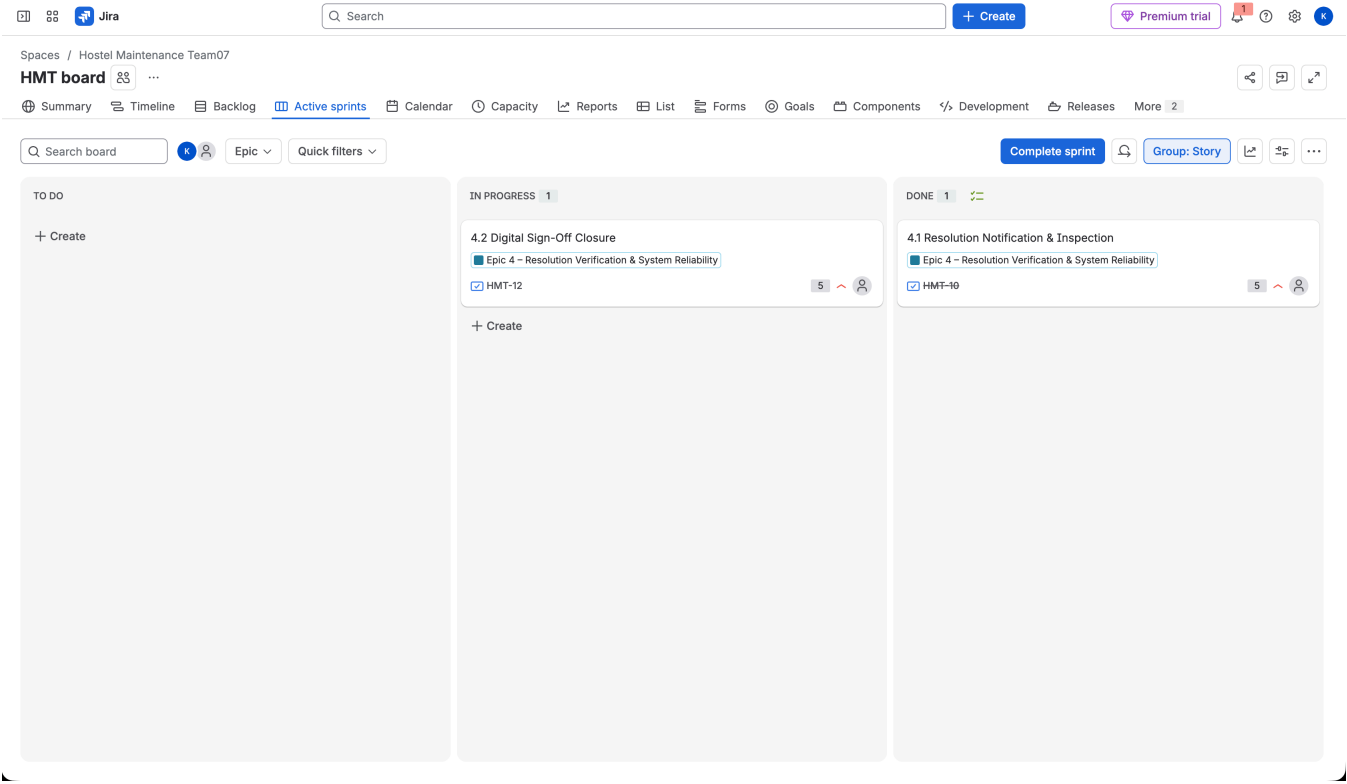


Figure 2: Sprint 1 Active Sprint board mid-sprint – Story 4.1 in DONE; Story 4.2 IN PROGRESS.

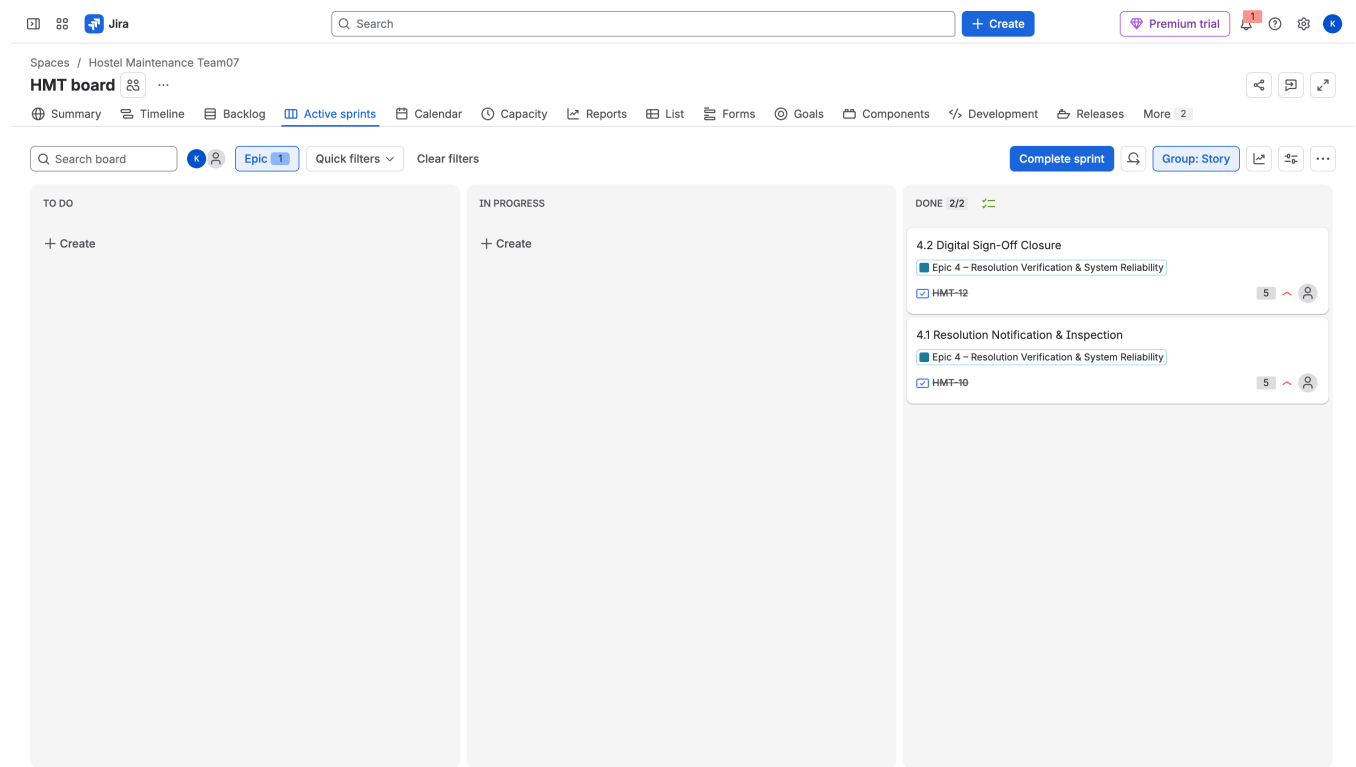


Figure 3: Sprint 1 Active Sprint board – all stories (10 pts) in DONE, before clicking Complete sprint.

Sprint 2 outcome

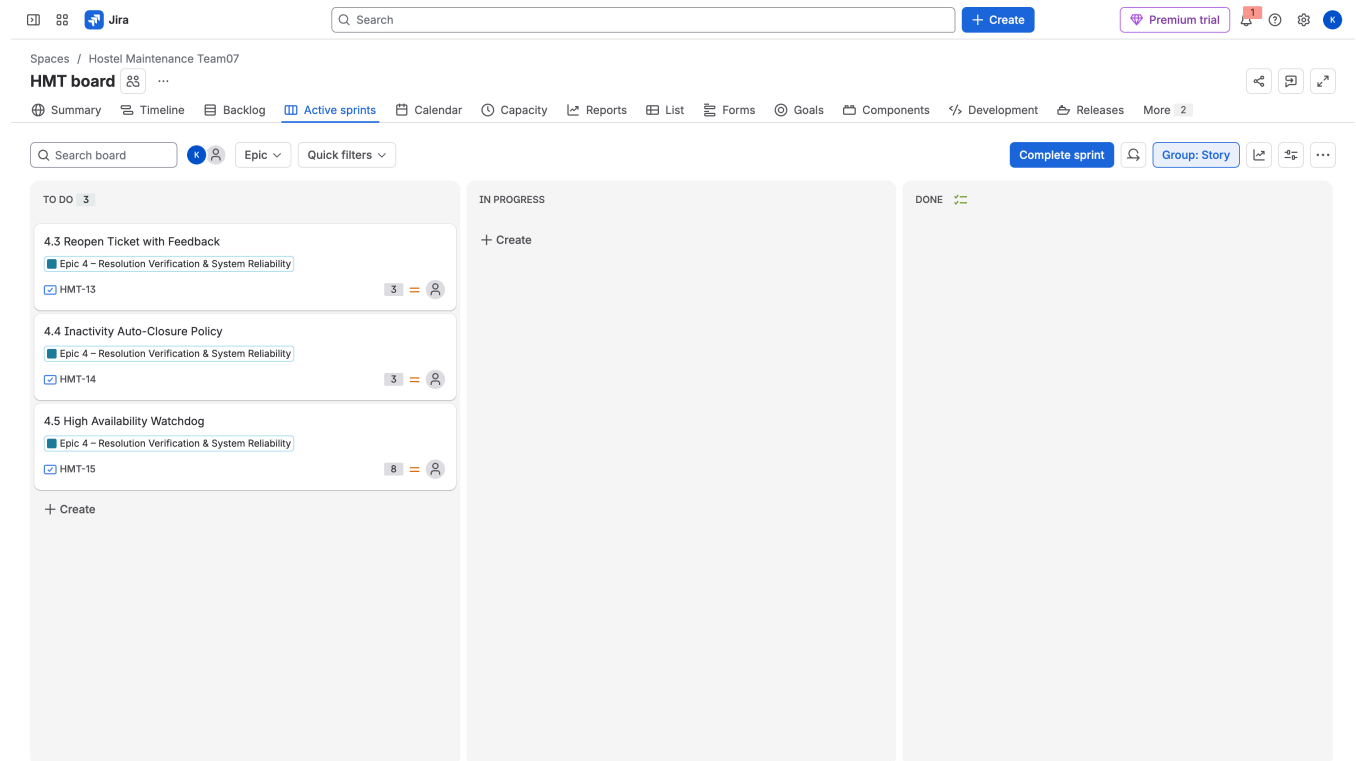


Figure 4: Sprint 2 Active Sprint board – all stories (14 pts) in TO DO.

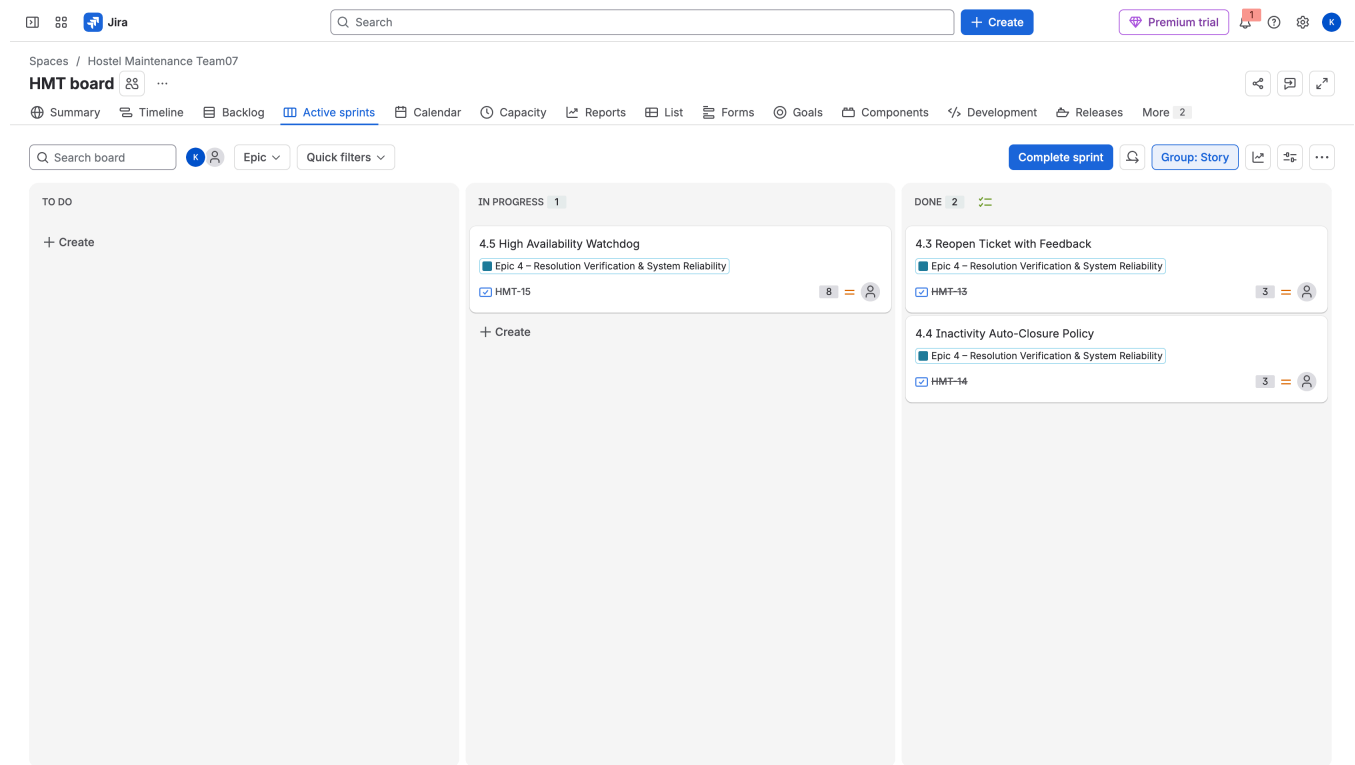


Figure 5: Sprint 2 Active Sprint board mid-sprint – Story 4.3 and 4.4 in DONE; Story 4.5 IN PROGRESS.

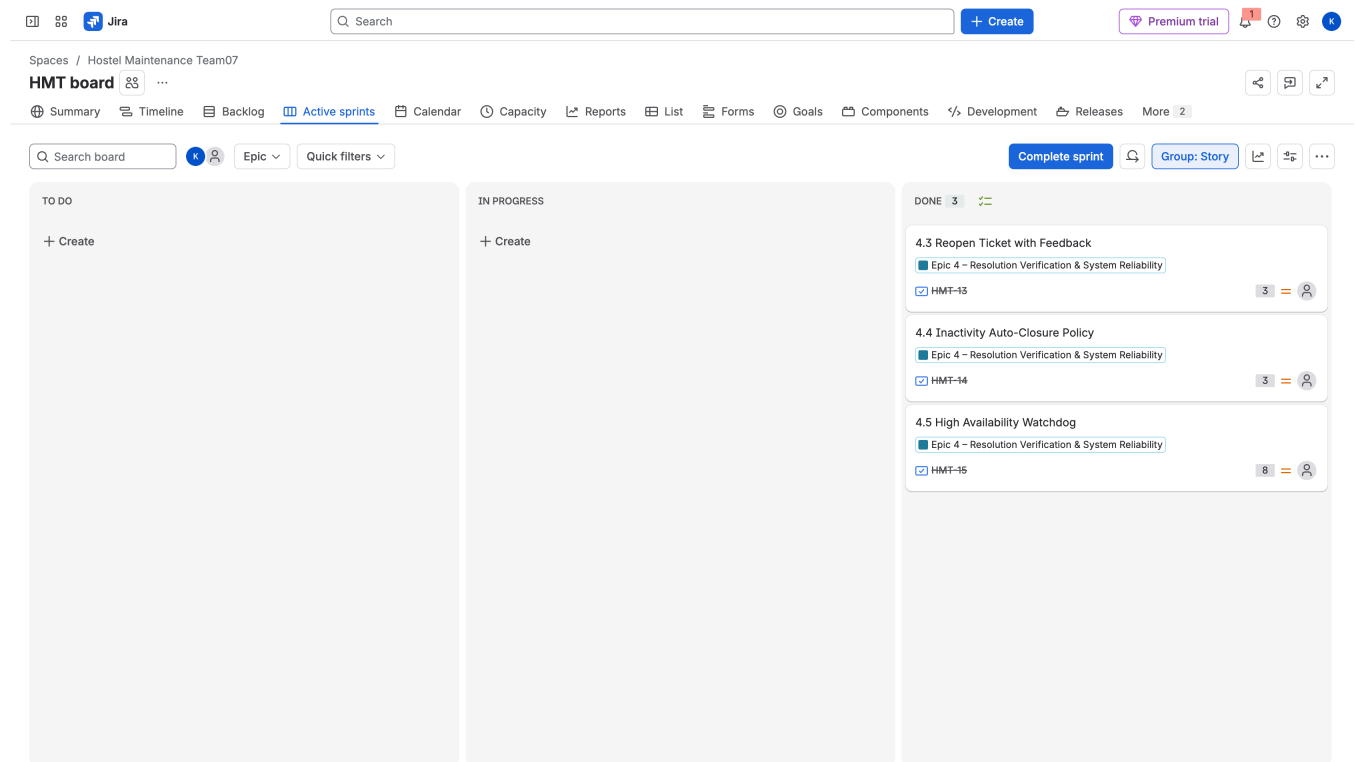


Figure 6: Sprint 2 Active Sprint board – all stories (14 pts) in DONE, before clicking Complete sprint.

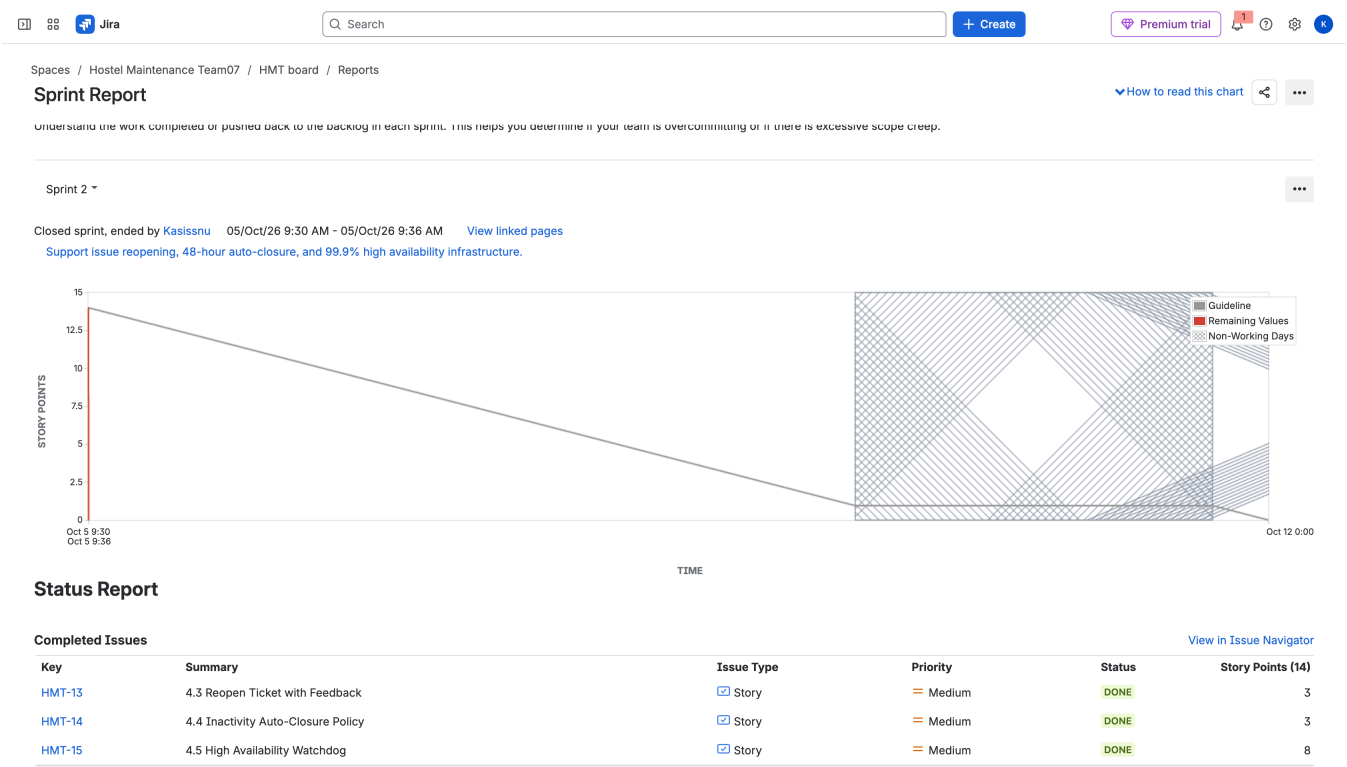


Figure 7: Sprint 2 outcome (Sprint Report, taken after completion) – all stories DONE, 14 story points.

7. Burndown Chart Analysis

The Jira Burndown Chart plots story points remaining (red line) against the ideal guideline (grey line). Each sprint is shown twice: once with non-working days displayed (hatched) and once with them hidden, which makes the guideline easier to read. The event table under each chart lists the exact completion times.

Sprint 1 Burndown (10 pts)

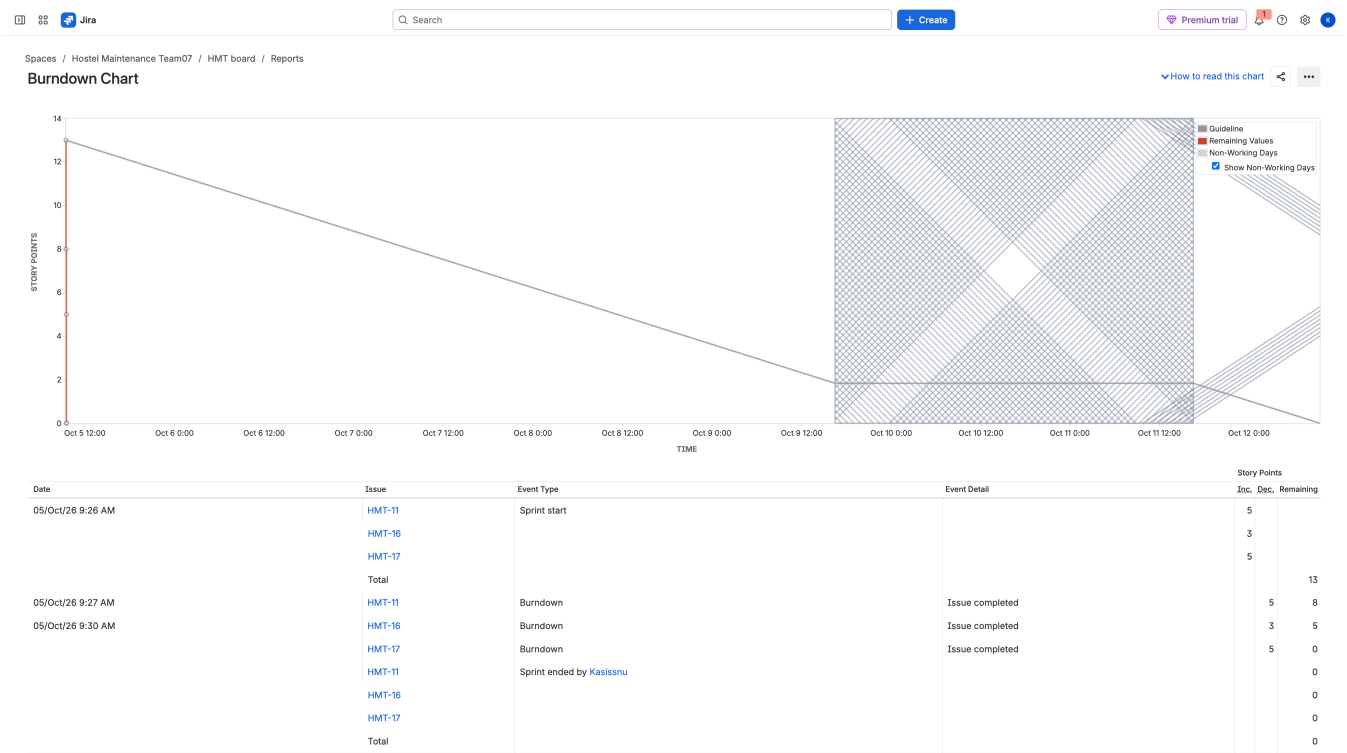


Figure 8: Sprint 1 Burndown, non-working days shown. 10 → 0.

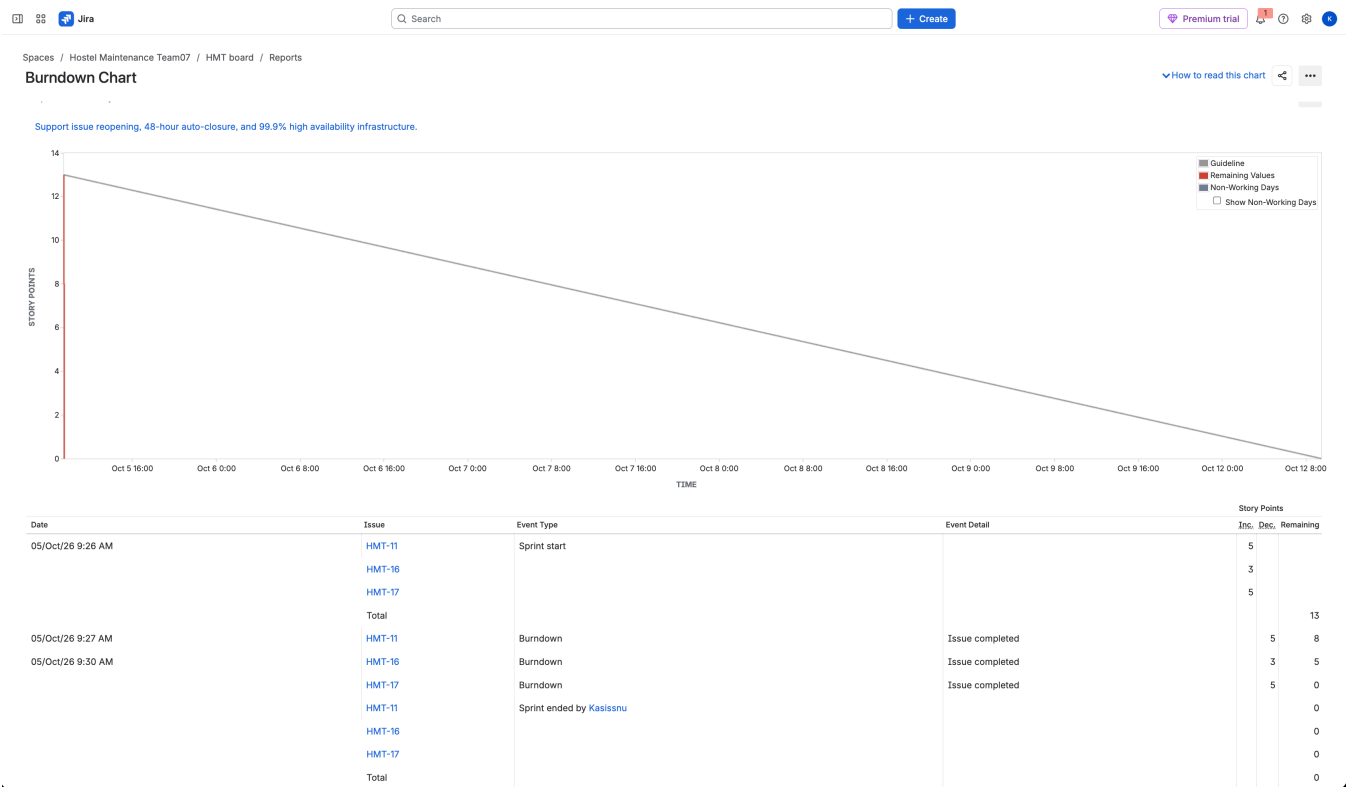


Figure 9: Sprint 1 Burndown, non-working days hidden.

Sprint 2 Burndown (14 pts)

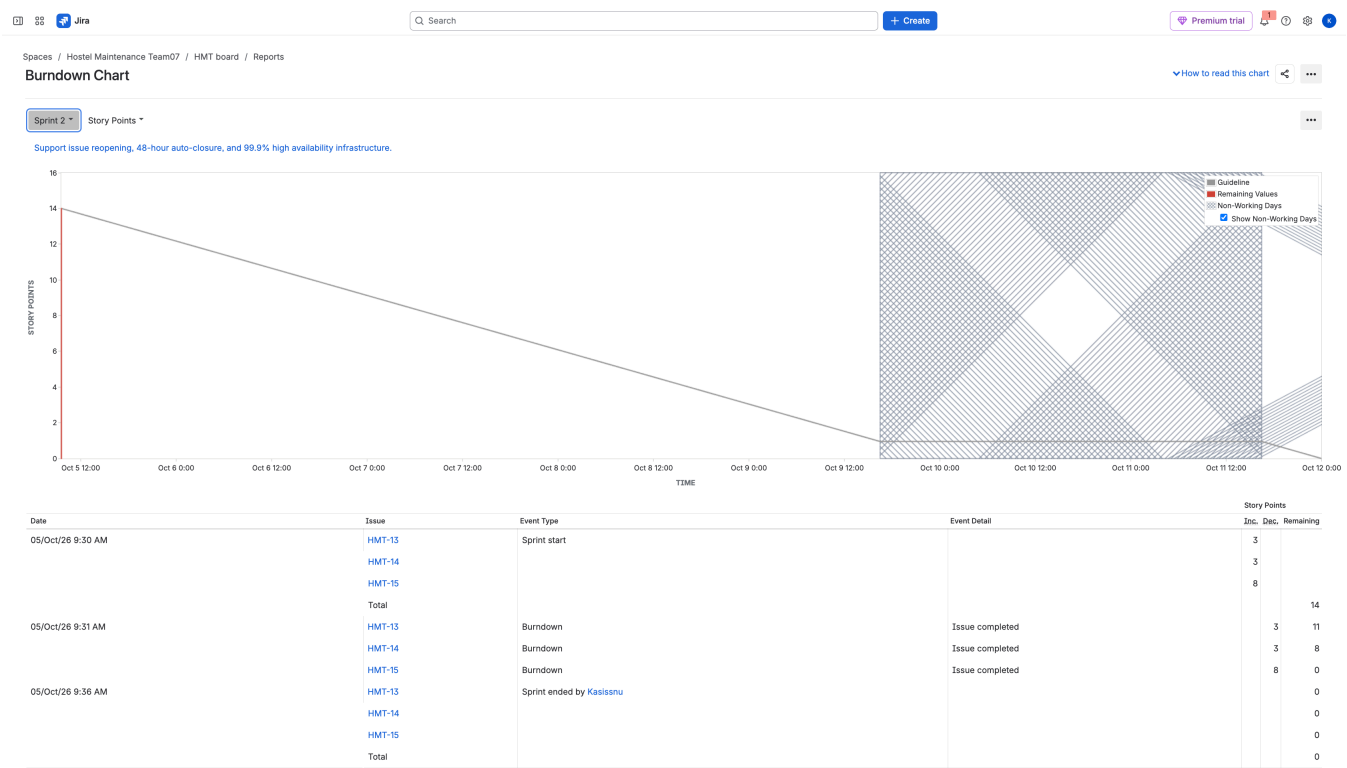


Figure 10: Sprint 2 Burndown, non-working days shown. 14 → 0.

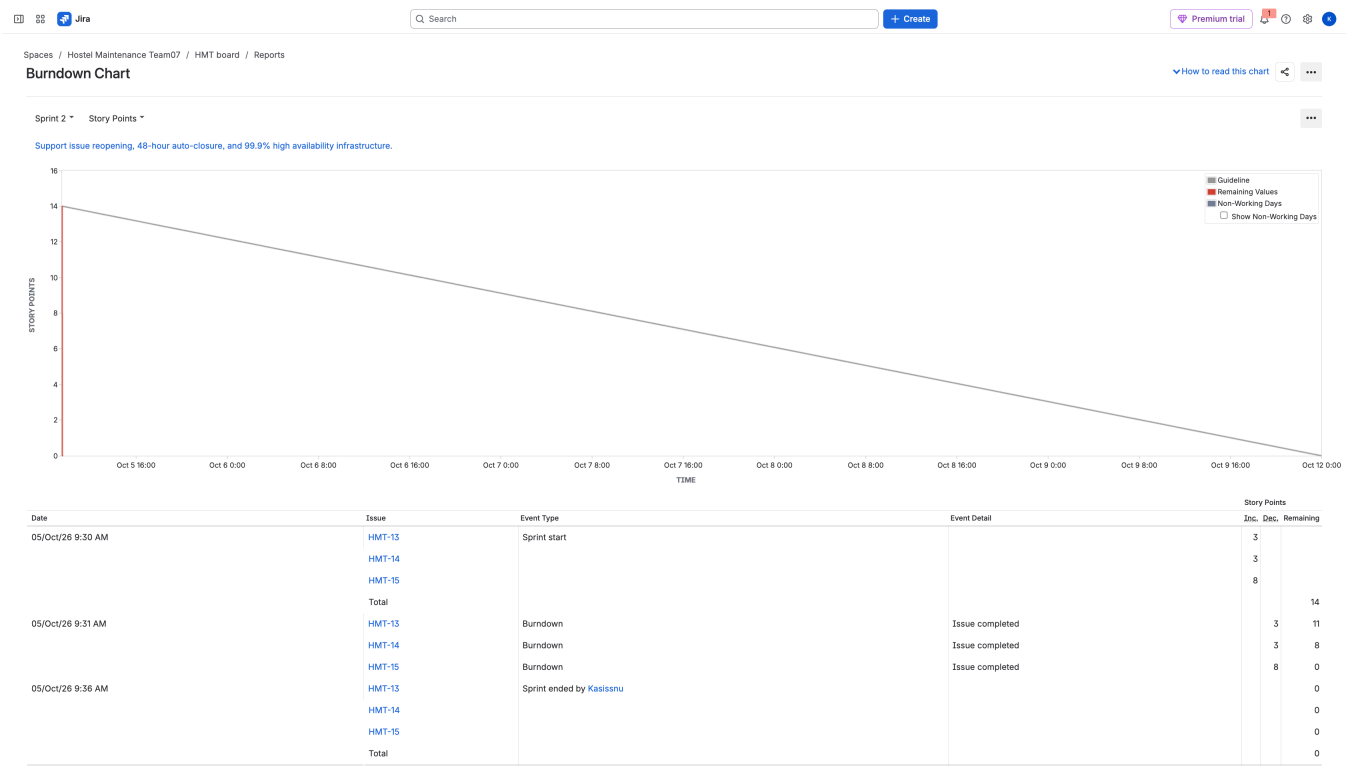


Figure 11: Sprint 2 Burndown, non-working days hidden.

Summary of the burndown data (from the Jira event tables)

Sprint	Start (points)	Completion events	End (points)
Sprint 1	6:45 PM – 10	6:48 PM HMT-21 (-5) → 5; 6:51 PM HMT-22 (-5) → 0	0
Sprint 2	7:05 PM – 14	7:08 PM HMT-23 (-3) → 11; 7:10 PM HMT-24 (-3) → 8; 7:14 PM HMT-25 (-8) → 0	0

8. Reflection Questions**1. Did your estimations reflect the actual effort?**

Yes. The 8-point estimate assigned to High Availability Watchdog correctly anticipated the complexity of infrastructure health monitoring, automated failover triggers, and uptime metric logging. Conversely, the digital sign-off button (5 points) and auto-closure timer (3 points) were straightforward database updates.

2. Was your backlog well-prioritized?

Yes. Sprint 1 targeted the primary happy path: resident sign-off and ticket closure. Sprint 2 targeted exceptions and reliability: the rework loop (reopen ticket), abandonment safeguards (48-hour timeout), and platform-wide 99.9% uptime monitoring.

3. How did your simulated sprint align with your plan?

Both sprints delivered 100% of planned story points (10 points in Sprint 1, 14 points in Sprint 2) with zero rollover, meeting both sprint goals.

4. What insights did the burndown chart give about your team's capacity?

The burndown revealed that committing 10 points in Sprint 1 and 14 points in Sprint 2 fell comfortably within realistic team velocity. The large step-down from the 8-point reliability watchdog showed that infrastructure stories should ideally be decomposed into smaller increments.